Statistical Analysis Discussion

November 6, 2026

Outline

1. "tximport"
2. ANOVA Basics
3. Negative Binomial GLM Basics
4. Interpreting $\sim$ `treatment * time * temperature`

"tximport" or not?

- My proposal: Just round the salmon decimals and be done with it.
- What does tximport do? Round it also but take into account different isoform usage (?)

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ANOVA Basics: Dummy coding

Consider 2 treatments and n observations.

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{n_1} \\ y_{n_1+1} \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_{n_1} \\ \epsilon_{n_1+1} \\ \vdots \\ \epsilon_n \end{bmatrix}$$

The parameter value β_1 is the **change with respect to basline (control)** β_0. It will have an assigned p-value, which corresponds (under certain conditions) to a simple t-test.

ANOVA Basics: What is a Design Matrix $\mathbf{X}$?

Definition

The design matrix $\mathbf{X}$ is a structured array that translates categorical factors and continuous variables into numerical predictors for a linear model ($\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$).

Example: 3 Treatments (T_1 ref, T_2, T_3) $\times$ 2 Time Points (t_1 ref, t_2) with interaction:

$$\begin{bmatrix} y_{T1,t1} \\ y_{T1,t2} \\ y_{T2,t1} \\ y_{T2,t2} \\ y_{T3,t1} \\ y_{T3,t2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_{T2} \\ \beta_{T3} \\ \beta_{\text{time}} \\ \beta_{T2:\text{time}} \\ \beta_{T3:\text{time}} \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \\ \epsilon_6 \end{bmatrix}$$

Columns: Intercept — T_2 dummy — T_3 dummy — Time dummy — $T_2 \times$ Time — $T_3 \times$ Time

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Count Data & The Negative Binomial Model

The Nature of RNA-Seq Count Data

Read counts (K_{ij}) are **non-negative integers** representing gene expression levels (sum of all fragments mappped to a gene).

- **Why not standard Linear Regression?** Assumes continuous, normally distributed errors, which fails for zero-heavy, highly skewed count data.
- **Why not Poisson?** Poisson assumes variance = mean. Real biological replicates show **overdispersion** (variance > mean).
- **The Solution (Negative Binomial):** Extends Poisson by adding a gene-specific dispersion parameter (α_i) to account for biological and technical variability:

$$\text{Var}(K_{ij}) = \mu_{ij} + \alpha_i \mu_{ij}^2$$

Generalized Linear Model (GLM) for K_{ij}

Definition of terms:

- K_{ij}: Read count for gene i in sample j ($K_{ij} \in \mathbb{N}_0$)
- s_{ij}: Size factor: **accounting for sequencing depth differences** (Median of Ratios normalization!!)
- μ_{ij}: Expected mean parameter of the Negative Binomial ($\mu_{ij} \in \mathbb{R}^+$)

The GLM Framework:

$$K_{ij} \sim \text{NB}(\text{mean} = \mu_{ij}, \text{dispersion} = \alpha_i)$$

$$\mu_{ij} = s_{ij} q_{ij}$$

The dispersion parameter α_i is estimated also, in a very smart way pooling information from all genes. We do not get into that, we assume it constant.

The negative binomial model for counts

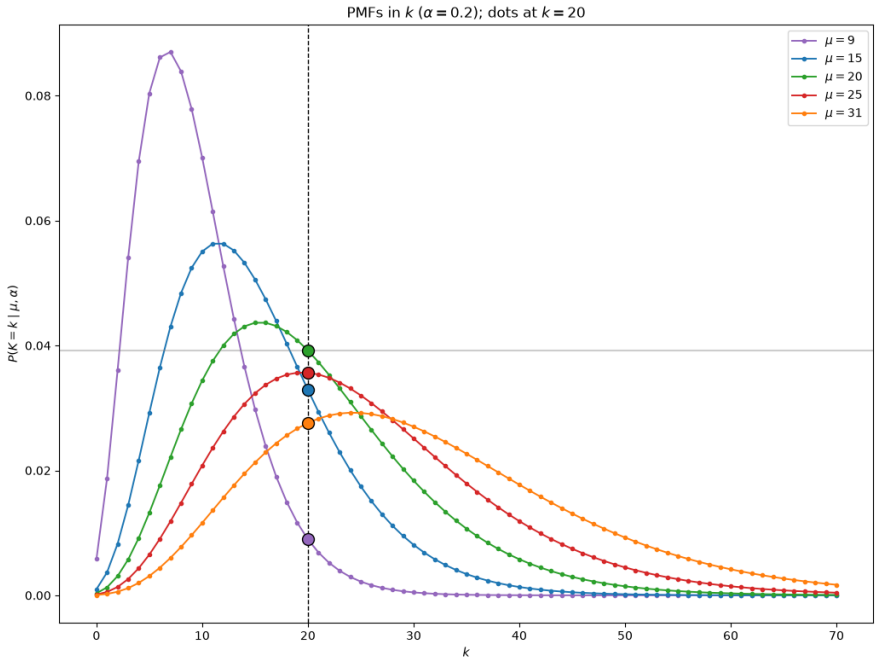
Read count of gene i in sample j:

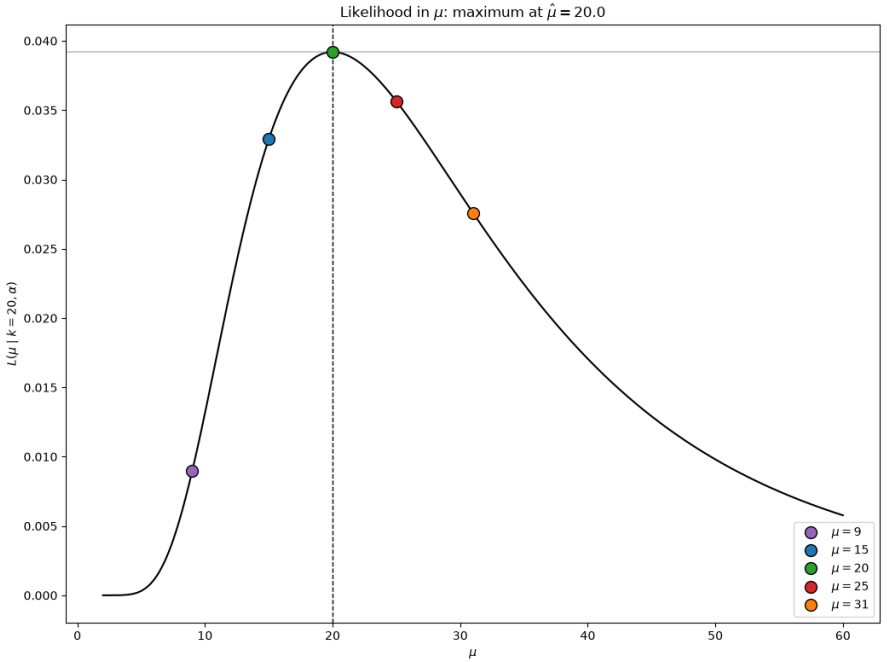
$$K_{ij} \sim \mathrm{NB}\left(\mu_{ij}, \alpha_i\right)$$

Probability mass function (scary looking, but you don't need to care about it):

$$P(K_{ij} = k) = \binom{k + 1/\alpha_i - 1}{k} \left(\frac{1}{1 + \alpha_i \mu_{ij}}\right)^{1/\alpha_i} \left(\frac{\alpha_i \mu_{ij}}{1 + \alpha_i \mu_{ij}}\right)^{k}$$

What we care about: μ is the expected count in each group


PMFs in k ($\alpha = 0.2$); dots at $k = 20$
$P(K = k \mid \mu, \alpha)$
k
$\mu = 9$
$\mu = 15$
$\mu = 20$
$\mu = 25$
$\mu = 31$


Likelihood in μ: maximum at $\hat{\mu} = 20.0$
$L(\mu \mid k = 20, \alpha)$
μ
$\mu = 9$
$\mu = 15$
$\mu = 20$
$\mu = 25$
$\mu = 31$

From ANOVA to counts: the DESeq2 model

Count of gene i in sample j: $K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$ with

$$\mu_{ij} = s_j q_{ij} \iff \frac{\mu_{ij}}{s_j} = q_{ij} \qquad \text{and} \qquad \log_2 q_{ij} = \sum_r x_{jr}\beta_{ir} = \mathbf{x}_j^\top \boldsymbol{\beta}_i$$

Depth-Normalized Abundance

$$\log_2 q_{ij} = \log_2\left(\frac{\mu_{ij}}{s_j}\right)$$

Interpretation: This represents the expected abundance of gene i in sample j, and it is the part that directly depends on your experiment design matrix X.

We are "predicting" the $\log_2$ of the depth normalized counts!

From ANOVA to counts: the DESeq2 model

$$\begin{bmatrix} \log_2 q_{i1} \\ \vdots \\ \log_2 q_{in_1} \\ \log_2 q_{i,n_1+1} \\ \vdots \\ \log_2 q_{in} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \beta_{i0} \\ \beta_{i1} \end{bmatrix}$$

Control: $q_{ij} = 2^{\beta_{i0}}$ baseline normalized count

Treated: $q_{ij} = 2^{(\beta_{i0}+\beta_{i1})} = 2^{\beta_{i0}}\, 2^{\beta_{i1}}$ baseline $\times$ fold change

$$\log_2 \mathrm{FC} = \log_2 \frac{q_{\text{treated}}}{q_{\text{control}}} = \log_2 \frac{2^{\beta_{i0}+\beta_{i1}}}{2^{\beta_{i0}}} = \log_2 \frac{2^{\beta_{i0}}\, 2^{\beta_{i1}}}{2^{\beta_{i0}}}$$

$$= \log_2 2^{\beta_{i1}} = \beta_{i1}$$

$\beta_{i1} = 1$: the normalized count doubles; $\beta_{i1} = -1$: it halves.

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Interpreting `treatment * time * temperature`: setup

Reference (baseline) condition

mock, $t = 0.5$, $21°C$

Each factor becomes a 0/1 indicator **relative to the reference**:

- $T = 1$: treated ($0 =$ mock)
- $t = 1$: later time ($0 = 0.5$)
- $H = 1$: other temperature ($0 = 21°C$)

The model has 8 coefficients (intercept, 3 main effects, 3 two-way and 1 three-way interaction):

$$\log_2 q_{ij} = \beta_0 + \beta_T T + \beta_t t + \beta_H H + \beta_{Tt}\, Tt + \beta_{TH}\, TH + \beta_{tH}\, tH + \beta_{TtH}\, TtH$$

Every coefficient is a $\log_2$ change relative to a specific condition.

Baseline and main effects

β_0 $\log_2$ normalized count in the **reference** (mock, 0.5, 21°C)

β_T treatment LFC vs mock, **only** at $t = 0.5$ and 21°C

β_t time effect, **only** in mock at 21°C

β_H temperature effect, **only** in mock at $t = 0.5$

Warning

With interactions in the model, a "main effect" is the effect **at the reference level of the other factors**. It is not the overall effect of the treatment.

Interactions: how an effect changes

β_{Tt} **change in the treatment LFC** at the later time (at 21°C)

β_{TH} **change in the treatment LFC** at the other temperature (at $t = 0.5$)

β_{tH} change in the time effect at the other temperature (in mock)

β_{TtH} change in β_{Tt} at the other temperature (a "difference of differences")

Treatment vs mock in each condition

In the treated-minus-mock difference, every term without T cancels, exactly as in the β_{i1} derivation:

$$\log_2 \mathrm{FC}_{(t,H)} = \beta_T + \beta_{Tt}\, t + \beta_{TH}\, H + \beta_{TtH}\, tH$$

Time	Temperature	$\log_2$ FC (treatment vs mock)
0.5	21°C	β_T
later	21°C	$\beta_T + \beta_{Tt}$
0.5	other	$\beta_T + \beta_{TH}$
later	other	$\beta_T + \beta_{Tt} + \beta_{TH} + \beta_{TtH}$

Only the first row is a single coefficient. The others are sums.